

- Overview
- Generation Mix
- Consumption

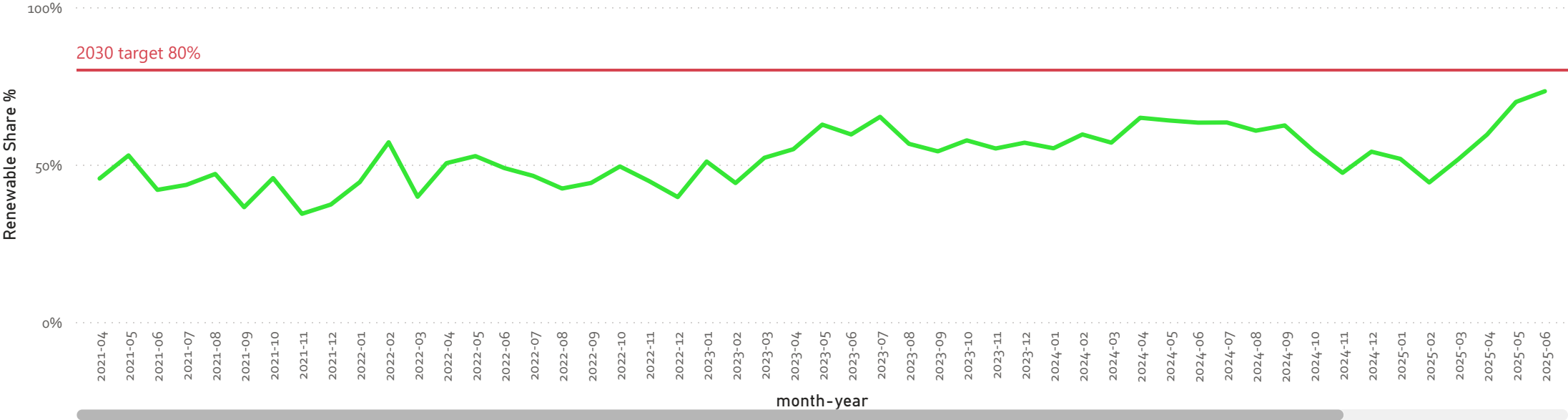
Germany's Energy Transition: Are We On Track for 2030?

Germany's renewable share has climbed from 47% in early 2021 to a peak of ~80% in summer 2025, narrowing the 2030 gap. But seasonal swings like solar-rich summers vs fossil-heavy winters reveal the real challenge isn't generation, but it's storage and grid flexibility.

filter by date

1/1/20214/1/2026

Germany's Renewable Energy Share Path to 2030



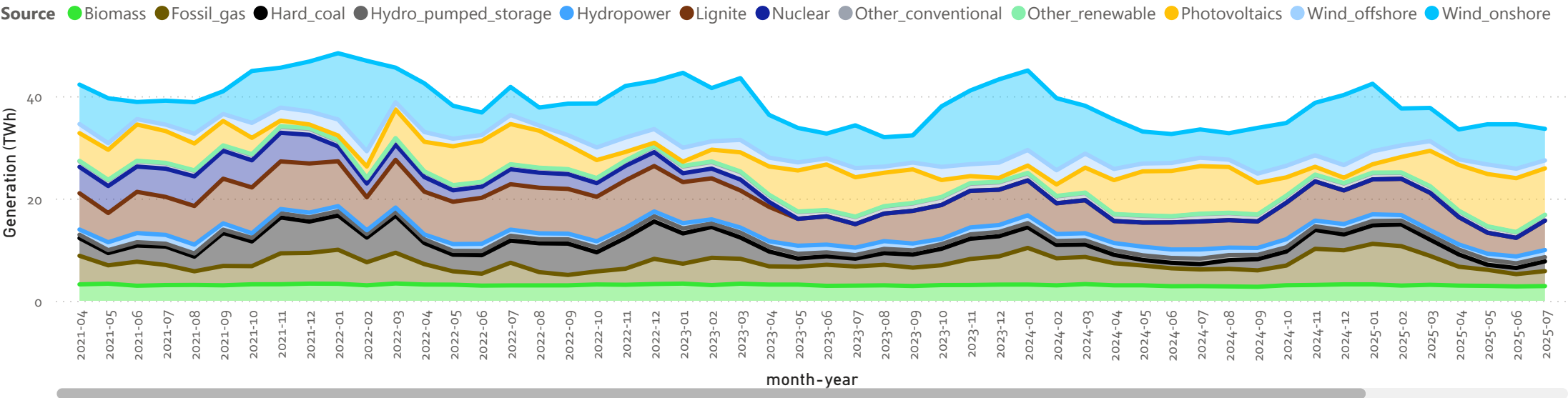
2.33K

Total Generation (TWh)

52.6%

Renewable Share %

Generation Mix by Source How the Energy Transition Looks



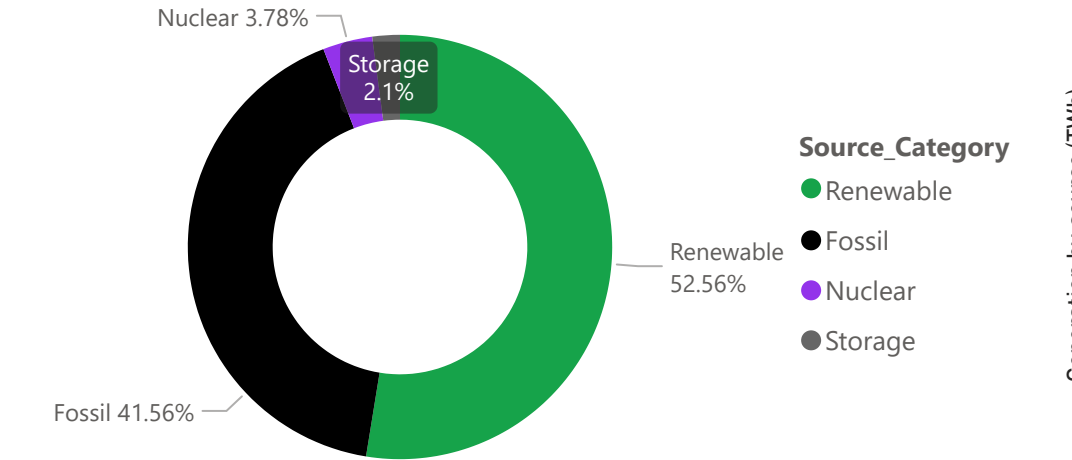
1.7%

Renewable Share YoY (pp)

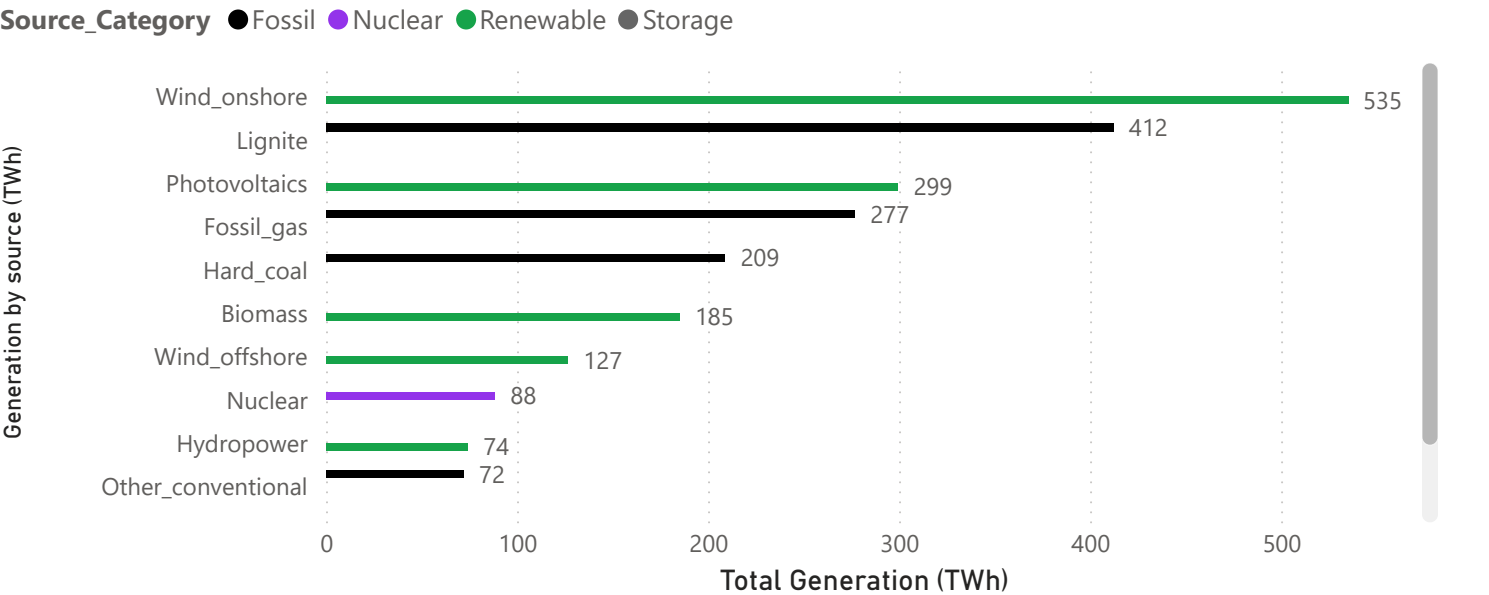
2.37K

Total Consumption (TWh)

Generation mix by category(5-year total)



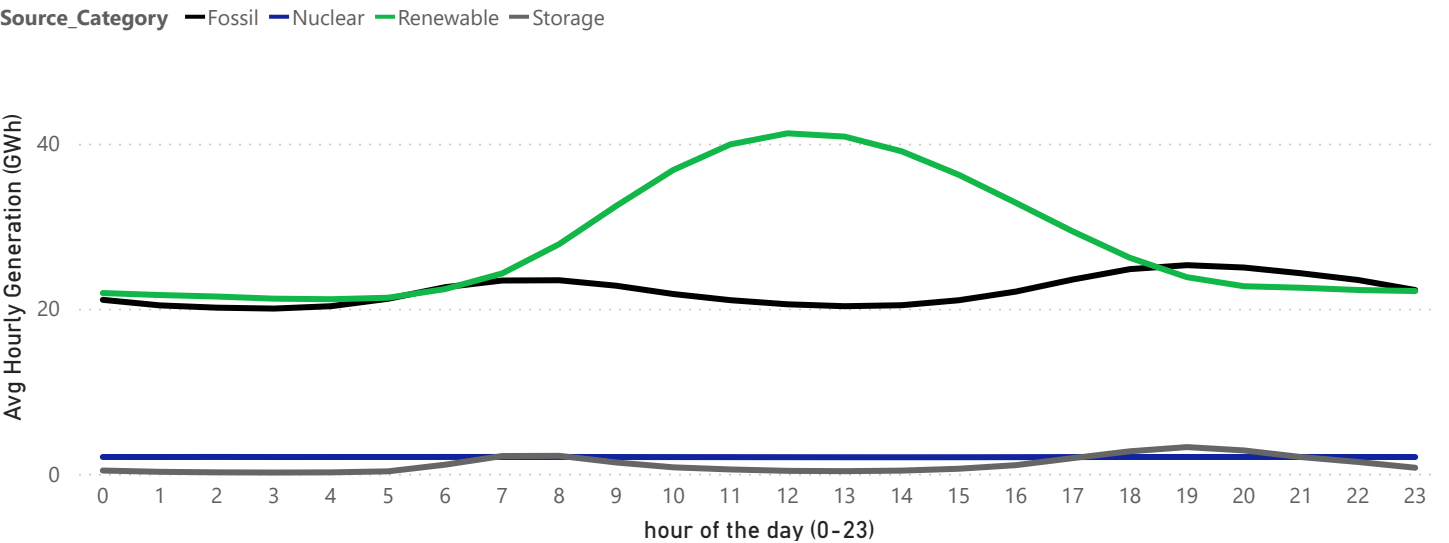
Total Generation (TWh) by Source and Source_Category



Generation Trend: Source × Year (TWh)

Source	2021	2022	2023	2024	2025	2026	Total
Biomass	28.55	37.74	37.30	36.20	35.87	9.16	184.81
Fossil_gas	39.87	45.78	51.09	56.92	60.55	22.51	276.70
Hard_coal	39.88	62.89	39.75	27.34	28.16	10.70	208.71
Hydro_pumped_storage	6.02	10.15	10.18	10.39	9.89	2.23	48.86
Hydropower	11.73	12.80	15.02	17.55	14.18	2.86	74.16
Lignite	72.41	103.53	77.84	70.99	67.17	20.13	412.06
Nuclear	48.64	32.82	6.74	0.00	0.00	0.00	88.20
Other_conventional	9.72	11.52	16.36	15.84	14.37	3.97	71.77
Other_renewable	1.00	1.29	1.21	1.06	0.95	0.22	5.73
Photovoltaics	39.58	55.73	55.54	63.15	73.78	11.51	299.30
Wind_offshore	16.65	24.75	23.52	25.67	26.19	9.73	126.50
Wind_onshore	62.87	100.99	119.01	112.56	106.77	33.10	535.29
Total	376.90	499.98	453.55	437.65	437.90	126.12	2,332.10

Average Daily Generation Profile by Hour (GWh)



Overview

Generation Mix

Consumption

The Demand Side: When Germany Actually Uses Electricity

Germany's consumption peaks at 11:00 with industrial demand. Solar generation peaks one hour later at 12:00. They almost align, but consumption stays elevated through 18:00, while solar drops off sharply after 14:00. That mismatched afternoon shoulder is where storage and grid flexibility matter most.

filter by date

1/1/2021

4/1/2026

2.37K

Total Consumption (TWh)

61.52

Peak Hourly Consumption (GWh)

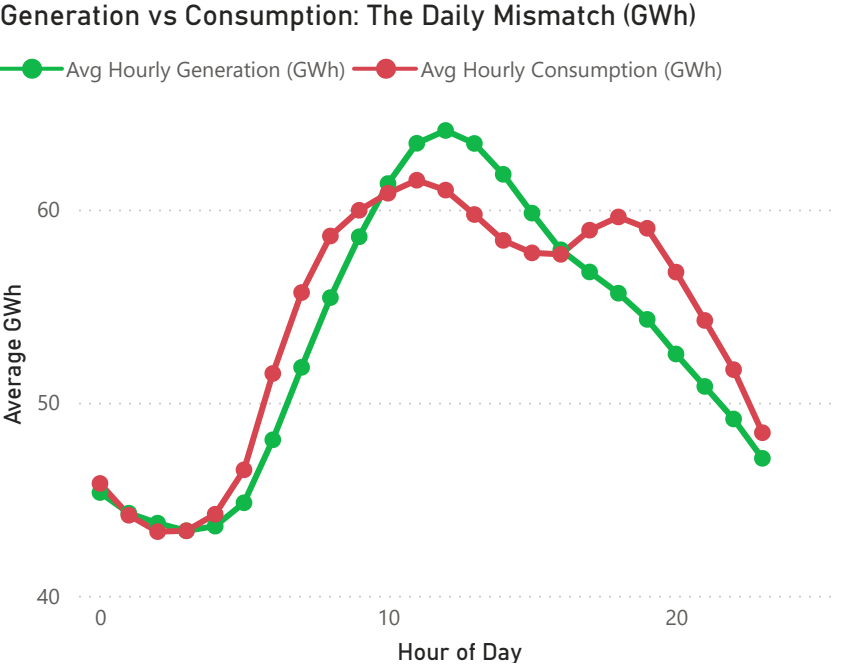
39.65

Generation Deficit (TWh)

54.12

Avg Hourly Consumption (GWh)

Day_of_Week	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	Total
Monday	43.4	42.1	41.5	41.8	43.2	46.6	53.7	58.9	62.0	63.0	63.6	64.2	63.8	62.8	61.6	60.8	60.5	61.6	62.2	61.6	59.2	56.4	53.4	49.9	55.7
Tuesday	47.1	45.5	44.7	44.9	46.1	49.2	56.0	61.0	63.7	64.3	64.8	65.3	64.8	63.8	62.5	61.8	61.5	62.5	63.0	62.4	59.9	57.0	53.8	50.3	57.3
Wednesday	47.5	45.9	45.2	45.4	46.5	49.6	56.2	61.2	63.8	64.5	65.0	65.5	64.9	63.9	62.6	61.9	61.6	62.6	63.1	62.5	60.0	57.1	54.0	50.4	57.5
Thursday	47.6	46.0	45.1	45.3	46.4	49.3	55.7	60.5	63.3	64.0	64.6	65.1	64.6	63.5	62.2	61.5	61.2	62.3	62.7	62.1	59.7	56.8	53.8	50.2	57.2
Friday	47.5	45.8	45.0	45.2	46.2	49.1	55.2	60.1	62.9	63.7	64.2	64.5	63.8	62.2	60.6	59.8	59.7	60.6	60.8	59.9	57.3	54.4	51.6	48.3	56.2
Saturday	45.6	43.7	42.5	42.2	42.3	42.7	44.1	46.7	50.1	52.7	54.1	54.9	54.3	52.7	51.4	50.9	51.2	52.7	53.7	53.0	50.9	48.9	47.2	44.5	48.9
Sunday	42.0	40.2	39.2	38.9	38.9	39.1	39.7	41.6	44.7	47.7	49.6	51.3	51.0	49.3	48.0	47.5	48.1	50.2	51.8	51.6	50.4	49.1	48.1	45.5	46.0
Total	45.8	44.2	43.3	43.4	44.2	46.5	51.5	55.7	58.6	60.0	60.8	61.5	61.0	59.7	58.4	57.8	57.7	58.9	59.6	59.0	56.8	54.3	51.7	48.4	54.1



Monthly Generation vs Consumption (TWh)

